

Topic 11: Organic chemistry - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Hydrocarbons, fuels, functional groups and polymers

Questions in this answer PDF: 62

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2020 Feb/March Paper 42 Question 1

Main topic: Topic 11: Organic chemistry | Marks: 13 | Answer source: 0620_m20_ms_42.pdf

Answer points

1(a)

- distillation (1)
- fractional (1)

1(b)

- naphtha (1)
- bitumen (1)

1(c)(i)

- any two from:
- methane ethane propane

1(c)(ii)

- $2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$
- CO_2 and H_2O as products (1)
- correct equation (1)

1(c)(iii)

- carbon monoxide

1(d)(i)

- kerosene is more viscous
- gasoline is more flammable

1(d)(ii)

- kerosene has molecules with a longer carbon chain

1(e)

- hydrogen + oxygen $\rightarrow$ water

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2020 Feb/March Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 8 | Answer source: 0620_m20_ms_42.pdf

Answer points

5(a)(i)

- monomer

5(a)(ii)

- any hydrocarbon with one $C=C$ bond (with both C atoms having 4 bonds) (1)
- structure of methylpropene (1)

5(b)

- water is a product (when polymer is made)

5(c)(i)

- any correct amide link between any two blocks showing all atoms and all bonds (1)
- correct orientation of 3 inter-block amide links (1)
- continuation bonds (1)

5(c)(ii)

- nylon

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

2020 May/June Paper 41 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 17 | Answer source: 0620_s20_ms_41.pdf

Answer points

5(a)

- sugar(s)
- fermentation
- ethene
- hydration

5(b)

- (acidified) potassium manganate(VII)

5(c)(i)

- (Liebig) condenser

5(c)(ii)

- arrow at the lower inlet

5(d)(i)

- methanal
- C₄H₈O

5(d)(ii)

- C_nH_{2n}O

5(e)(i)

- functional group

5(e)(ii)

- 4 x C-H dot cross bonds and 1 C-C dot cross bond
- 1 x C=O dot cross bond
- non-bonding electrons on O

5(f)(i)

- (structural) isomers

5(f)(ii)

- any structure with correct valencies and formula of C₃H₆O
- C=O bond on second carbon (of a chain of 3)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

2020 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 16 | Answer source: 0620_s20_ms_42.pdf

Extra topic(s): Topic 5: Chemical energetics, Topic 2: Atoms, elements and compounds, Topic 7: Acids, bases and salts

Answer points

6(a)(i)

- substitution

6(a)(ii)

- ultraviolet light / ultraviolet radiation

6(a)(iii)

6(b)(i)

- only one product / double bond becomes single bond / two molecules join (to make one molecule)

6(b)(ii)

- (energy required to break bonds =) 854 (1)
- (energy given out when bonds form =) 1025 (1)
- overall energy change 854 - 1025 = -171 (1)

6(c)(i)

- alkene (1)
- alcohols / alkanols (1)
- carboxylic acids/alkanoic acids (1)

6(c)(ii)

- turns colourless / decolourised (1)
- bubbles / fizzing / effervescence (1)

6(d)(i)

- C-C (1)
- each C bonded to -CH₂OH / -COOH and H with no other atoms + extension bonds (1)

6(d)(ii)

- polyester
- C
- C
- COOH
- CH₂OH
- H
- H

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2020 May/June Paper 43 Question 1

Main topic: Topic 11: Organic chemistry | Marks: 10 | Answer source: 0620_s20_ms_43.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

1(a)(i)

- D

1(a)(ii)

- C

1(a)(iii)

- C

1(a)(iv)

- B

1(a)(v)

- D AND E or A

1(a)(vi)

- C

1(a)(vii)

- or E

1(a)(viii)

- C
- C
- C
- H
- H
- H
- H
- H
- H
- O
- H
- H

1(b)

- donate or lose protons
- does not fully ionise / partially ionises / forms an equilibrium mixture

Easy explanation / reminder

- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2020 May/June Paper 43 Question 8

Main topic: Topic 11: Organic chemistry | Marks: 6 | Answer source: 0620_s20_ms_43.pdf

Answer points

8(a)(i)

8(a)(ii)

8(a)(iii)

- water

8(b)

- non-biodegradable / running out of space / toxic / leaching

8(c)

- (complex) carbohydrate / protein

Easy explanation / reminder

- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

2020 Oct/Nov Paper 41 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 16 | Answer source: 0620_w20_ms_41.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 2: Atoms, elements and compounds

Answer points

6(a)(i)

- alkanes

6(a)(ii)

- one mark each for any two of:

- -

- same chemical properties

- -

- same functional group

- -

- same general formula

- -

- (consecutive members) differ by CH₂

- -

- common (allow similar) methods of preparation

- -

- physical properties vary in predictable manner / show trends / gradually change OR example of a physical property

- variation i.e. melting point / boiling point / volatility (1)

6(b)

- 3.01×10^{23} (molecules)

6(c)(i)

- HCl

6(c)(ii)

6(c)(ii)

- substitution

6(d)(i)

- from: orange (1)

- to: colourless (1)

6(d)(ii)

- contains no double bonds/ethane only contains single bonds

6(e)(i)

- (1)

- (1)

6(e)(ii)

- C₂H₄

6(f)(i)

- (C = 85.7, H = 14.3, Mr 112)

- C = 85.7

- $12 = 7.14$ H = 14.3

- = 14.3 (1)

- (ratio = $7.13 : 14.3 = 1 : 2$)

- CH₂ (2)

6(f)(ii)

- C₈H₁₆

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2020 Oct/Nov Paper 41 Question 7

Main topic: Topic 11: Organic chemistry | Marks: 8 | Answer source: 0620_w20_ms_41.pdf

Answer points

7(a)(i)

- sugar or C₆H₁₂O₆, is renewable / sustainable

7(a)(ii)

- slow(er) process

7(b)(i)

- ethanoic acid

7(b)(ii)

- oxidation

7(c)(i)

- ethyl methanoate

7(c)(ii)

7(d)

- ethanol: (forces of attraction) between molecules (1)
- sodium chloride: (force of attraction) between positive and negative ions/ionic bonding (1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2020 Oct/Nov Paper 42 Question 4

Main topic: Topic 11: Organic chemistry | Marks: 10 | Answer source: 0620_w20_ms_42.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds

Answer points

4(a)

- hydrocarbon

4(b)(i)

- (only) one product is formed

4(b)(ii)

- 3 single bonded C atoms
- entire molecule correct

4(b)(iii)

- orange (1) to colourless (1)

4(b)(iv)

- structure of but-1-ene
- structure of but-2-ene

4(c)

- C₃H₈ + Cl₂ → C₃H₇Cl + HCl
- HCl as product
- rest of equation

Easy explanation / reminder

- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2020 Oct/Nov Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 22 | Answer source: 0620_w20_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

5(a)(i)

- presence of an acid

5(a)(ii)

- oxidation

5(a)(iii)

- all single bonding dot and cross pairs correct (1)
- double C=O bond dot and cross pairs are correct (1)
- Complete diagram is correct (1)

5(b)

- partial (dissociation) (1)
- full / 100% (dissociation) (1)
- both acid colours (1)
- HCl indicating a lower pH acid colour than CH₃COOH (1)
- fizzing OR dissolving / disappearing in both (1)
- either observation happens quicker with HCl (1)

5(c)(i)

- water

5(c)(ii)

- acid (catalyst)

5(c)(iii)

- any one alcohol group on a molecule (1)
- structure of propan-2-ol (1)

5(d)(i)

- the atoms / group of atoms which give (any molecule its) chemical properties

5(d)(ii)

- C₃H₅O₂

5(d)(iii)

- Mr of HO-CH₂-CH₂-OH = 62 (1)
- %ge = $100 \times 32/62 = 52\%$ (1)

5(d)(iv)

- any correct ester link between any two blocks showing all atoms and all bonds (1)
- correct orientation of 3 inter-block ester links and continuation bonds (1)

5(d)(v)

- Terylene

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2020 Oct/Nov Paper 43 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 14 | Answer source: 0620_w20_ms_43.pdf

Answer points**6(a)(i)**

- compounds with the same molecular formula (1)
- different structural formula (1)

6(a)(ii)

- diagram (1)
- diagram (1)
- ethanoic acid (1)
- methyl methanoate (1)

6(b)(i)**6(b)(ii)**

- condensation

6(b)(iii)

- 1 for each correct functional group

6(c)

- one correct linkage fully displayed (1)
- the whole structure fully correct (1)

6(d)(i)

- proteins

6(d)(ii)

- amino acids

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2021 Feb/March Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 20 | Answer source: 0620_m21_ms_42.pdf

Answer points**5(a)**

- R and U only

5(b)

- aqueous bromine (1)
- turns colourless / decolourises (1)

5(c)(i)

- but-2-ene

5(c)(ii)

- structure of methyl propene

5(d)(i)

- C₁₂H₂₆

5(d)(ii)

- P, R and U

5(e)

- steam (1)
- catalyst (1)
- One other condition (1):
- either 60 atm or 300 °C

5(f)(i)

- oxidation

5(f)(ii)

- potassium manganate(VII)

5(g)(i)

- ester

5(g)(ii)

- name = methanol
- structure =
- name = butanoic acid structure =

5(g)(iii)

- C₅H₁₀O₂

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 Feb/March Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 9 | Answer source: 0620_m21_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points**6(a)**

- monomers

6(b)(i)

- any hydrocarbon with one C=C bond (with both C atoms having 4 bonds) (1)
- structure of but-2-ene (1)

6(b)(ii)

- addition

6(c)(i)

- protein

6(c)(ii)

- any correct amide link between any two blocks showing all atoms and all bonds (1)
- identical orientation of both inter-block amide links including terminal groups with correct orientation (if shown) (1)
- continuation bonds on polymer (1)

6(c)(iii)

- hydrolysis (1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2021 May/June Paper 41 Question 7

Main topic: Topic 11: Organic chemistry | Marks: 15 | Answer source: 0620_s21_ms_41.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points**7(a)**

- 48.65 / 12 8.11 / 1 43.24 / 16 (1)
- OR evaluation
- 4.05 8.11 2.7(0)
- divide all by smallest
- OR 1.5 : 3 : 1
- OR 6 : 3 : 2 (1)
- C₃H₆O₂ (1) ALLOW symbols in any order

7(b)

- (Mr of CH₄O = 32)
- CH₄O (1)

7(c)(i)

- C_nH_{2n}O₂
- OR
- C_nH_{2n+1}COOH

7(c)(ii)

- butanoic acid (1)
- fully displayed carboxylic acid group (1)
- correct structure of butanoic acid showing all atoms and bonds (1)

7(c)(iii)

- homologous series

7(d)(i)

- brown to colourless

7(d)(ii)

- C₉H₂₀ (1)
- 2C₃H₆ (1)

7(d)(iii)

- addition

7(d)(iv)

- any one repeat unit (1)
- both repeat units fully correct (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 12 | Answer source: 0620_s21_ms_42.pdf

Answer points**6(a)(i)**

- carboxylic acid

6(a)(ii)

- any correct displayed ester link between any two blocks showing all atoms and all bonds (1)
- correct orientation of three displayed inter-block ester links with correct orientation (1)

- continuation bonds on polyester (1)

6(a)(iii)

- water

6(b)(i)

- two blocks linked by -O-

6(b)(ii)

- acid (and heat) (1)
- enzymes (ignore names) (1)

6(b)(iii)

- chromatography

6(c)(i)

- fermentation

6(c)(ii)

- $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
- C_2H_5OH (1)
- rest of equation (1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2021 May/June Paper 43 Question 1

Main topic: Topic 11: Organic chemistry | Marks: 7 | Answer source: 0620_s21_ms_43.pdf

Extra topic(s): Topic 4: Electrochemistry

Answer points

1(a)

- polymerisation

1(b)

- fractional distillation

1(c)

- fermentation

1(d)

- distillation

1(e)

- electrolysis

1(f)

- chromatography

1(g)

- filtration
- Question Answer

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 May/June Paper 43 Question 7

Main topic: Topic 11: Organic chemistry | Marks: 15 | Answer source: 0620_s21_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

7(a)

- 11.63 / 1 69.77 / 12 18.6 / 16
- Or evaluation
- 11.63 5.81 1.16
- Or ratio
- 10 : 5 : 1 (1)
- $C_5H_{10}O$ (1)

7(b)

- (M_r of CH_2O =) 30 (1)
- $C_2H_4O_2$ (1)

7(c)(i)

- structural isomers

7(c)(ii)

- carboxylic acids
- Question Answer

7(c)(iii)

- name of ester, corresponding to $C_3H_6O_2$: ethyl methanoate or methyl ethanoate (1)
- correctly displayed ester linkage (1)
- fully correct displayed formula corresponding to $C_3H_6O_2$ and matching named ester (1)

7(c)(iv)

- displayed formula of propanoic acid (1)
- propanoic acid (1)

7(d)

- Step 1:
- cracking (1)
- $C_{10}H_{22} \rightarrow C_2H_4 + C_8H_{18}$ (1)
- Step 2:
- (catalytic) addition (1)
- $C_2H_4 + H_2O \rightarrow C_2H_5OH$ (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 Oct/Nov Paper 41 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 15 | Answer source: 0620_w21_ms_41.pdf

Answer points**5(a)**

- homologous series

5(b)

- C_nH_{2n}

5(c)(i)

- orange to colourless

5(c)(ii)

- (only) one product is formed

5(c)(iii)

- $C_4H_8 + Br_2 \rightarrow C_4H_8Br_2$
- $C_4H_8Br_2$ (1)
- equation fully correct (1)

5(c)(iv)

- (poly) but-2-ene

5(c)(v)

- but-1-ene (1)
- structure of but-1-ene
- (1)

5(d)(i)

- C_2H_4O

5(d)(ii)

- $2C_3H_7COOH + Na_2CO_3 \rightarrow 2C_3H_7COONa + H_2O + CO_2$
- C_3H_7COONa (1)
- equation fully correct (1)

5(d)(iii)

- methyl butanoate (1)
- ester link (1)
- rest of structure (1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2021 Oct/Nov Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 12 | Answer source: 0620_w21_ms_42.pdf

Answer points

6(a)

- $C_nH_{2n+1}OH$ (1)

6(b)

- not all atoms are C or H

6(c)(i)

- C_2H_4

6(c)(ii)

- glucose

6(d)

- $C_2H_5OH + 3O_2 \rightarrow 3H_2O + 2CO_2$
- species (1) correct equation (1)

6(e)(i)

- $C_3H_5O_2$

6(e)(ii)

- any one ester link (1)
- rest of molecule (1)

6(e)(iii)

- water

6(f)(i)

- carboxylic acid

6(f)(ii)

- structure of ethanedioic acid

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2021 Oct/Nov Paper 43 Question 7

Main topic: Topic 11: Organic chemistry | Marks: 16 | Answer source: 0620_w21_ms_43.pdf

Answer points

7(a)

- 1 for each for any 2 of:
 - -
 - (same) general formula
 - -
 - similar chemical properties
 - -
 - (consecutive members) differ by CH_2
 - -
 - same functional group
 - -
 - common (allow similar or same) methods of preparation
 - -
 - physical properties and vary in predictable manner or show trends or gradually change
 - OR example of a physical property variation, e.g.
 - melting points increase
 - boiling points increase
 - volatility decreases

7(b)(i)

- cracking

7(b)(ii)

- $\rightarrow C_2H_4 + C_4H_{10}$ (2)

7(b)(iii)

- $C_2H_4 + H_2O \rightarrow C_2H_5OH$

7(b)(iv)

- hydration / addition

7(b)(v)

- 1 mark for ANY 1 of:

--

- 300 °C

--

- 60 atmospheres

--

- H₃PO₄

7(c)

- 1 mark each for any 2 of:

--

- carbohydrates are renewable

--

- fossil fuels are non-renewable OR fossil fuels conserved

--

- lower temperature OR lower energy

--

- hydration is equilibrium meaning lower yield

--

- lower pressure (used)

7(d)(i)

- substitution

7(d)(ii)

- addition

7(d)(iii)

- any one repeat unit with extension bonds (1)

- both units fully correct (1)

7(e)

- 1 mark for each functional group fully correct (2)

Easy explanation / reminder

- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2022 Feb/March Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 11 | Answer source: 0620_m22_ms_42.pdf

Answer points

5(a)(i)

- cracking

5(a)(ii)

- C₁₀H₂₂ → 4C₂H₄ + C₂H₆

- C₁₀H₂₂ as only reactant

- formulae of ethene and ethane as only products

- correct equation

5(b)(i)

- hydrogen chloride

5(b)(ii)

- ultraviolet light

5(c)(i)

- (only) one product is formed

5(c)(ii)

- Bond energy in breaking bonds

- = [(4 × 410) + 610 + 240] = 2490 (kJ / mol)

- Use of total E change to find bond energy of C₂H₄Cl₂

- = + 180 = 2490 + 180 = 2670 (kJ / mol)

- Determination of total C-Cl bond energy

- = - [(4 × 410) + 350] = 2670 - 1990 = 680 (kJ / mol)

- Determination of each C-Cl bond energy

- = / 2 = 680 / 2 = 340 (kJ / mol)

Easy explanation / reminder

- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.
- Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.

2022 Feb/March Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 17 | Answer source: 0620_m22_ms_42.pdf

Answer points

6(a)(i)

- homologous series

6(a)(ii)

- Same functional group

6(a)(iii)

- general (formula)

6(a)(iv)

- February/March 2022

6(b)

- fractional
- distillation

6(c)(i)

- propanoic acid
- structure of propanoic acid

6(c)(ii)

- methanol

6(d)(i)

- structural isomers

6(d)(ii)

- ethyl butanoate
- butyl ethanoate

6(e)(i)

- any correct displayed ester link between any two blocks showing all atoms and all bonds
- correct orientation of three displayed inter-block ester links with correct orientation
- continuation bonds on polyester

6(e)(ii)

- condensation

6(e)(iii)

- terylene

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

2022 May/June Paper 41 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 19 | Answer source: 0620_s22_ms_41.pdf

Answer points

5(a)

- the rate of forward reaction equals the rate of the reverse reaction (1)
- concentrations of reactants and products are constant (1)

5(b)

- reaction too slow (1)
- yield of ethanoic acid too low (1)

5(c)

- increases (1)
- decreases (1)
- decreases (1)

5(d)

- cobalt (1)
- transition element (1)

5(e)

- $C_nH_{2n+1}COOH$

5(f)

- COOH fully displayed (1)
- whole molecule completely correct (1)

5(g)(i)

- methyl butanoate

5(g)(ii)

- butanoic acid (1)
- methanol (1)

5(h)

- C 48.65 / 12
- H 8.11 / 1
- O 43.24 / 16
- OR
- 4.05:8.11:2.70 (1)
- fractions shown dividing all by smallest
- OR
- 1.5:3:1
- OR
- 3:6:2 (1)
- $C_3H_6O_2$ (1)

5(i)

- $C_4H_8O_2$

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2022 May/June Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 15 | Answer source: 0620_s22_ms_42.pdf

Answer points**5(a)**

- fermentation

5(b)

- (complete) combustion
- $C_2H_5OH + 3O_2 \rightarrow 3H_2O + 2CO_2$
- reactant species
- fully correct equation

5(c)(i)

- ethene

5(c)(ii)

- addition

5(c)(iii)

- steam
- catalyst + 60 atm or catalyst + 300 °C

5(d)(i)

- hydrogen
- catalyst
- high temperature

5(d)(ii)

- C_nH_{2n+2}

5(e)(i)

- potassium manganate(VII)

5(e)(ii)

- carboxylic acids

5(e)(iii)

- fully displayed formula of ethanoic acid

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2022 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 9 | Answer source: 0620_s22_ms_42.pdf

Answer points

6(a)(i)

6(a)(ii)

- diagram of dioic acid showing two fully displayed carboxylic acid groups on unshaded block
- diagram of diol showing two fully displayed hydroxyl groups on shaded block

6(a)(iii)

- polyester

6(b)

- one

6(c)(i)

- any hydrocarbon with one C=C bond (with both C atoms having 4 bonds) (1)
- structure of but-2-ene (1)
- but-2-ene

6(c)(ii)

- cracking

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2022 May/June Paper 43 Question 4

Main topic: Topic 11: Organic chemistry | Marks: 18 | Answer source: 0620_s22_ms_43.pdf

Answer points

4(a)

- (a substance which) increases the rate of a reaction
- remains unchanged or unaffected or without changing mass

4(b)

- increases (1)
- decreases (1)
- decreases (1)
- decreases (1)

4(c)(i)

- $C_nH_{2n+1}OH$

4(c)(ii)

- any 2 from:
 - ■
 - same or similar chemical properties or reactions
 - ■
 - (contain) the same functional group
 - ■
 - (consecutive members) differ by CH₂
 - ■
- physical properties vary in predictable manner / show trends / gradually change

4(c)(iii)

- (1)
- propan-2-ol (1)

4(d)(i)

- ethyl butanoate

4(d)(ii)

- ethanol (1)
- butanoic acid (1)

4(e)

- C 58.82 / 12 H 9.80 / 1 O 31.37 / 16
- OR
- 4.90:9.80:1.96 (1)
- 2.5:5:1
- OR
- 5:10:2 (1)
- C₅H₁₀O₂(1)

4(f)

- C₆H₁₂O₂

Easy explanation / reminder

- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*
- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2022 Oct/Nov Paper 41 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 9 | Answer source: 0620_w22_ms_41.pdf

Extra topic(s): Topic 5: Chemical energetics

Answer points

5(a)

- C_nH_{2n}

5(b)

- orange to colourless

5(c)

- addition

5(d)(i)

- horizontal line below energy level to right hand side of reactants line and labelled C₂H₄Br₂ (1)
- activation energy 'hump' with upward arrow labelled from the reactants level (1)
- one downward arrow starting from the energy level of the reactants and finishing at the energy level of the products (1)

5(d)(ii)

- energy needed to break bonds
- 4 -> C-H + C=C + Br-Br = 4 -> 410 + 610 + 190 = 2440 (kJ) (1)
- energy released in making bonds
- 4 -> C-H + C-C + 2 -> C-Br = 4 -> 410 + 350 + 2 -> 290 = 2570 (kJ) (1)
- energy change = - 130 (kJ / mol) (1)

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2022 Oct/Nov Paper 41 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 12 | Answer source: 0620_w22_ms_41.pdf

Answer points

6(a)(i)

- methyl methanoate

6(a)(ii)

- CH₂O

6(b)

- all single bonding dot and cross pairs correct (1)
- double C=O bond dot and cross pairs are correct (1)
- complete diagram is correct (1)

6(c)

- methanoic acid (1)
- structure of methanoic acid (1)
- methanol (1)
- structure of methanol (1)

6(d)(i)

- same (molecular) formula (1)

- different structural formula (1)

6(d)(ii)

- ethanoic acid

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2022 Oct/Nov Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 10 | Answer source: 0620_w22_ms_42.pdf

Extra topic(s): Topic 5: Chemical energetics

Answer points

5(a)

- C_nH_{2n+2}

5(b)

- needs ultraviolet (light)

5(c)

- substitution

5(d)

- hydrogen chloride

5(e)(i)

- activation energy

5(e)(ii)

- energy (change) of reaction

5(e)(iii)

- energy of products is lower than energy of reactants

5(f)

- energy needed to break bonds
- 6 → C-H + C-C + Cl-Cl = 6 → 410 + 350 + 240 = 3050 (kJ / mol) (1)
- energy released in making bonds
- 5 → C-H + C-C + C-Cl + H-Cl = 5 → 410 + 350 + 340 + 430 = 3170 (kJ / mol) (1)
- energy change = -120 (kJ / mol) (1)

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2022 Oct/Nov Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 12 | Answer source: 0620_w22_ms_42.pdf

Answer points

6(a)

- C_nH_{2n+1}COOH

6(b)

- methanoic acid

6(c)

- CH₂O₂

6(d)

- all single bonding dot and cross pairs correct (1)
- double C=O bond dot and cross pairs are correct (1)
- four non-bonding cross electrons on each O atom (1)

6(e)(i)

- propyl methanoate (1)
- ester linkage with all atoms and all bonds (1)
- correct structure of propyl methanoate (1)

6(e)(ii)

- water

6(e)(iii)

- methyl propanoate (1)

- butanoic acid (1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2022 Oct/Nov Paper 43 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 18 | Answer source: 0620_w22_ms_43.pdf

Extra topic(s): Topic 5: Chemical energetics

Answer points

6(a)(i)

- 2670 (1)
- 2610 (1)
- (+) 60 (1)

6(a)(ii)

- endothermic AND energy released when bonds form is less than energy absorbed to break bonds
- OR
- endothermic AND overall energy change has a positive sign

6(b)(i)

6(b)(ii)

- but-1-ene (1)
- C=C (both C atoms should have 4 bonds) (1)
- Displayed formula of but-1-ene (1)

6(b)(iii)

- CH₂

6(c)(i)

- hydrolysis

6(c)(ii)

- acids (1)
- enzymes (1)

6(c)(iii)

- chromatography (1)
- locating agent (1)
- R_f (1)

6(d)(i)

- nylon

6(d)(ii)

- carboxylic acid (1)
- amine (1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2023 Feb/March Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 17 | Answer source: 0620_m23_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

5(a)

- same functional group

5(b)

- carboxylic acids

5(c)

- C_nH_{2n+1}COOH

5(d)

- propan-2-ol (1)
- displayed formula of propan-2-ol (1)

5(e)(i)

- C₁₀H₂₂ → C₃H₈ + C₃H₆ + C₄H₈

- $C_3H_8 + C_3H_6$ (1)
- rest of equation correct (1)
- February/March 2023

5(e)(ii)

- cracking

5(f)(i)

- six C atoms joined by single bonds in a chain and with continuation bonds (1)
- 3 $\rightarrow$ CH₃ at two C intervals and whole structure correctly displayed (1)

5(f)(ii)

- addition

5(g)(i)

- sodium propanoate

5(g)(ii)

- CH₃CH₂COO-

5(h)(i)

- acid
- or
- any named inorganic acid, e.g. phosphoric acid

5(h)(ii)

- ethyl propanoate

5(h)(iii)

- displayed formula of any ester linkage (1)
- correct structure of ethyl propanoate (1)

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2023 May/June Paper 41 Question 1

Main topic: Topic 11: Organic chemistry | Marks: 6 | Answer source: 0620_s23_ms_41.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

1(a)

- B

1(b)

- A

1(c)

- H

1(d)

- J

1(e)

- G

1(f)

- D

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2023 May/June Paper 41 Question 7

Main topic: Topic 11: Organic chemistry | Marks: 12 | Answer source: 0620_s23_ms_41.pdf

Answer points

7(a)(i)

- needs or uses ultra violet light

7(a)(ii)

- displayed formula of 1-chlorobutane(1)
- displayed formula of 2-chlorobutane(1)

7(b)(i)

- C₄H₆O₃

7(b)(ii)

- alkene(1)
- alcohol(1)
- carboxylic acid(1)

7(b)(iii)

- turns colourless(1)
- bubbles / fizzing / effervescence(1)

7(b)(iv)

- only two carbon atoms joined by a single bond and two additional bonds on each(1)
-
- (1)

7(b)(v)

- polyester

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2023 May/June Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 11 | Answer source: 0620_s23_ms_42.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds, Topic 3: Stoichiometry

Answer points

5(a)(i)

- substitution

5(a)(ii)

- provide activation energy

5(a)(iii)

- photochemical

5(a)(iv)

- C₃H₈ + Cl₂ → C₃H₇Cl + HCl
- HCl equation correct

5(b)(i)

- it has a carbon-carbon bond which is not a single bond

5(b)(ii)

- CH₃CHClCH₂Cl

5(c)

- displayed formula of propan-1-ol
- displayed formula of propan-2-ol
- propan-1-ol
- (as either name)
- propan-2-ol (under displayed formula of propan-2-ol)

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2023 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 7 | Answer source: 0620_s23_ms_42.pdf

Answer points

6(a)

- ethyl butanoate

6(b)

- water

6(c)

- C₃H₆O

6(d)(i)

- 1 repeat unit circled

6(d)(ii)

- displayed diol on correct box (unshaded)
- displayed dioic acid on correct box (shaded)

6(d)(iii)

- condensation

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2023 May/June Paper 43 Question 1

Main topic: Topic 11: Organic chemistry | Marks: 6 | Answer source: 0620_s23_ms_43.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds

Answer points**1(a)**

- J

1(b)

- D

1(c)

- B

1(d)

- G

1(e)

- F

1(f)

- C

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2023 May/June Paper 43 Question 7

Main topic: Topic 11: Organic chemistry | Marks: 12 | Answer source: 0620_s23_ms_43.pdf

Answer points**7(a)(i)**

- ultraviolet (light)

7(a)(ii)

- displayed formula of
- 1-chloropropane
- displayed formula of 2-chloropropane

7(b)(i)

- any 2 from:
- (same) general formula
- (same / similar) chemical properties or reactions
- (contain the same) functional group

7(b)(ii)

- single bond between the two C atoms (and nothing on continuation bonds)
- 1 CH₃ group and 1 H on first carbon and 1 CH₃ group and 1 H on second carbon and whole structure correct

7(c)(i)

- structural formula of
- propan-1,3-dioic acid
- structural formula of
- ethan-1,2-diol

7(c)(ii)

- polyester

7(c)(iii)

- carboxylic acids
- alcohols

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

2023 Oct/Nov Paper 41 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 17 | Answer source: 0620_w23_ms_41.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

6(a)

- yeast (1)
- absence of air (1)

6(b)(i)

- 300 (1)
- 6000 (1)

6(b)(ii)

- ethene

6(b)(iii)

- only one product

6(c)(i)

- proton donor

6(c)(ii)

- partial dissociation

6(c)(iii)

- 4 → -2 or -8 (1)
- $P + (4 \rightarrow -2) = -3 \therefore P = +5$ (1)

6(d)

- uses renewable resources (1)
- high rate of reaction (1)

6(e)(i)

- potassium manganate(VII)

6(e)(ii)

- reducing agent

6(f)(i)

- calcium ethanoate

6(f)(ii)

- (CH₃COO)₂Ca

6(f)(iii)

- hydrogen

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

2023 Oct/Nov Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 11 | Answer source: 0620_w23_ms_42.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

6(a)

- (same) general formula (1)
- (contain the same) functional group (1)

6(b)

- -CH₂- unit

6(c)(i)

- alkenes (1)
- alkanes (1)
- carboxylic acids (1)

6(c)(ii)

- propanoic acid (1)
- displayed formula of propanoic acid (1)

6(d)(i)

- working to give the mass of atoms shown in Fig 6.1
- $103 - 14 - (2 \times 12) - (2 \times 16) - (4 \times 1)$
- (= 29)
- 'R' = C_2H_5

6(d)(ii)

- proteins

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2023 Oct/Nov Paper 43 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 12 | Answer source: 0620_w23_ms_43.pdf

Answer points**6(a)**

- any two
- -
- general formula
- -
- functional group
- -
- chemical properties

6(b)

- ethyl propanoate

6(c)(i)

- propan-1-ol (1)
- methanoic acid (1)

6(c)(ii)

- $\text{C}_4\text{H}_8\text{O}_2$

6(d)

- $\text{C}_2\text{H}_4\text{O}$

6(e)(i)

- condensation

6(e)(ii)

-
- 2 -COOH groups fully displayed (1)
-
- 2 -NH₂ groups fully displayed (1)

6(f)(i)

- proteins

6(f)(ii)

- amino acids

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2024 Feb/March Paper 42 Question 4

Main topic: Topic 11: Organic chemistry | Marks: 12 | Answer source: 0620_m24_ms_42.pdf

Extra topic(s): Topic 6: Chemical reactions, Topic 7: Acids, bases and salts

Answer points**4(a)(i)**

- structural

4(a)(ii)

- CH_2O_2

4(b)

- ethyl methanoate
- ester link

- correct displayed formula of ethyl methanoate

4(c)(i)

- nothing can enter or leave

4(c)(ii)

- the rate of forward reaction equals (the rate of the) reverse reaction
- concentrations of reactants and products are constant
- February/March 2024

4(c)(iii)

- increases
- decreases
- decreases
- no effect

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2024 Feb/March Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 15 | Answer source: 0620_m24_ms_42.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds, Topic 3: Stoichiometry

Answer points

5(a)

- (but-1-ene) has a lower relative molecular mass

5(b)

- carbon dioxide AND water

5(c)(i)

- photochemical

5(c)(ii)

- to provide the activation energy

5(c)(iii)

- substitution

5(c)(iv)

5(d)(i)

- acid

5(d)(ii)

- displayed formula of butan-1-ol
- displayed formula of butan-2-ol
- butan-1-ol
- butan-2-ol

5(e)(i)

- addition (polymerisation)
- February/March 2024

5(e)(ii)

- chain of six C atoms joined by single bonds in a chain
- three correctly placed C₂H₅ groups
- correct structure AND continuation bonds

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 May/June Paper 41 Question 7

Main topic: Topic 11: Organic chemistry | Marks: 13 | Answer source: 0620_s24_ms_41.pdf

Answer points

7(a)(i)

- C 82.76 / 12: H 17.24/1 (1)
- OR evaluation i.e.

- 6.90:17.24
- fractions showing division of both by smaller ie 6.9 / 6.9 and 17.24 / 6.9
- OR evaluation ie 1:2.5 OR 4:10
- C₂H₅ (1)

7(a)(ii)

- C₅H₁₀

7(b)

-

- any C=C with both carbons having a valency of 4(1)
- correct displayed formula of (cis or trans) but-2-ene(1)
- but-2-ene

7(c)

- C₈H₁₆

7(d)(i)

- structural isomers

7(d)(ii)

- ethanoic acid(1)
- methanol(1)

7(d)(iii)

- Displayed formula of propanoic acid showing ALL atoms and bonds
- (1)
- propanoic acid(1)

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2024 May/June Paper 42 Question 4

Main topic: Topic 11: Organic chemistry | Marks: 14 | Answer source: 0620_s24_ms_42.pdf

Answer points

4(a)

- C₂H₄O

4(b)

- methyl propanoate

4(c)

- displayed formula of propanoic acid
- propanoic acid

4(d)

- structure of CH₃OH
- methanol

4(e)

- decreases
- decreases
- decreases
- no effect

4(f)(i)

- 7
- 3 ■ pH ■ 7

4(f)(ii)

- H⁺(aq)

4(f)(iii)

- universal indicator

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2024 May/June Paper 42 Question 6

Answer points

6(a)(i)

- photosynthesis

6(a)(ii)

- $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

6(a)(iii)

- energy from light
- presence of chlorophyll

6(b)(i)

- fermentation

6(b)(ii)

- carbon dioxide

6(c)(i)

- 300 °C
- 60 atm or 6000 kPa
- acid

6(c)(ii)

- only one product is formed

6(c)(iii)

- C-C bond as dot and cross and C-O as 'cross and dot'
- all C-H bonds and O-H bond as dot and cross
- 4 non-bonding dot electrons on O atom and no other non-bonding electrons on C or H

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 May/June Paper 43 Question 7

Main topic: Topic 11: Organic chemistry | Marks: 14 | Answer source: 0620_s24_ms_43.pdf

Answer points

7(a)(i)

- C 83.33 / 12: H 16.67 / 1
- or
- 6.94:16.67
- C 6.94 / 6.94: H 16.67 / 6.94
- or 1:2.4
- or 5:12
- C_5H_{12}

7(a)(ii)

- Mr of $\text{C}_2\text{H}_5 = 29$
- $58 / 29 = 2$ and C_4H_{10}

7(b)(i)

- $\text{C}=\text{C}$

7(b)(ii)

-
- C
- C
- any $\text{C}=\text{C}$ with both carbons having a valency of 4
- displayed formula of
- but-1-ene
- but-1-ene

7(c)

- $2\text{C}_2\text{H}_4$
- C_8H_{18}

7(d)(i)

- $\text{C}_n\text{H}_{2n} + 1\text{OH}$

7(d)(ii)

- displayed formula of propan-1-ol or propan-2-ol
- propan-1-ol
- or
- propan-2-ol

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*

2024 Oct/Nov Paper 41 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 17 | Answer source: 0620_w24_ms_41.pdf

Answer points

5(a)(i)

- oxygen

5(a)(ii)

- advantage: any one of:
 -
 - water is the only product
 -
 - no carbon dioxide produced
 -
 - more efficient
- disadvantage: any one of:
 -
 - hydrogen needs to be stored at high pressure
 -
 - hydrogen hard to store
 -
 - heavy tanks needed to store hydrogen
 -
 - fewer (hydrogen) filling stations
 -
 - less efficient

5(b)(i)

- fractional distillation

5(b)(ii)

- cracking

5(c)

- Mr of CH₃CH₂COOCH₃ = 88
- mol of CH₃CH₂COOCH₃ = 44 / 88 = 0.5
- mol of H = 0.5 -> 8 = 4.0
- no of H atoms = 4.0 -> 6.02 -> 10²³ = 2.408 -> 10²⁴

5(d)

- ethane
- propene
- ethanol
- propanoic acid

5(e)(i)

- contain carbon and hydrogen atoms only

5(e)(ii)

- they have a carbon-carbon double bond

5(e)(iii)

- nickel

5(e)(iv)

- only one product is formed

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.

2024 Oct/Nov Paper 41 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 14 | Answer source: 0620_w24_ms_41.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

6(a)

- condensation

6(b)

- proteins

6(c)

- (3)
- NH₂ group displayed
- COOH group displayed
- use of R

6(d)

- correct amide link between any two blocks showing all atoms and all bonds
- identical orientation of inter-block amide links including terminal groups with correct orientation
- continuation bonds on polymer

6(e)

- distance travelled by substance
- distance travelled by solvent

6(f)(i)

- pencil is insoluble (in solvent)

6(f)(ii)

- locating agent

6(f)(iii)

- same R_f value

6(f)(iv)

- use a different solvent

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

2024 Oct/Nov Paper 42 Question 1

Main topic: Topic 11: Organic chemistry | Marks: 9 | Answer source: 0620_w24_ms_42.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

1(a)

- G

1(b)

- H

1(c)

- B

1(d)

- D

1(e)

- E

1(f)

- B

1(g)

- A

1(h)

- H

1(i)

- G

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2024 Oct/Nov Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 11 | Answer source: 0620_w24_ms_42.pdf

Answer points

5(a)

- petroleum

5(b)

- high temperature
- catalyst

5(c)(i)

- $C_{12}H_{26} \rightarrow 3C_4H_8 + H_2$
- C_4H_8
- correct equation

5(c)(ii)

- thermal decomposition

5(d)(i)

- poly(propene)

5(d)(ii)

- chain of six C atoms joined by single bonds in a chain
- three correctly placed CH_3 groups
- correct structure AND continuation bonds

5(d)(iii)

- addition

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2024 Oct/Nov Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 14 | Answer source: 0620_w24_ms_42.pdf

Answer points

6(a)(i)

- one amide linkage circled

6(a)(ii)

- two fully displayed carboxylic acid groups on the shaded box
- two fully displayed amino groups on the unshaded box

6(a)(iii)

- water

6(a)(iv)

- protein(s)

6(a)(v)

- amino acids

6(a)(vi)

- displayed formula of 2-aminopropanoic acid (3)
- displayed NH_2 or displayed $COOH$ group present
- displayed NH_2 and displayed $COOH$ groups attached to same C atom
- correct displayed structure of 2-aminopropanoic acid

6(b)(i)

- dicarboxylic acids
- diols

6(b)(ii)

- any one correct fully displayed ester link between any two blocks
- three fully displayed inter-block ester links with correct orientation
- fully correct diagram showing two complete repeat units of polyester including continuation bonds

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

2024 Oct/Nov Paper 43 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 9 | Answer source: 0620_w24_ms_43.pdf

Answer points

6(a)

- compound containing carbon and hydrogen only

6(b)(i)

- $\text{C}_{12}\text{H}_{26}$

6(b)(ii)

- cracking

6(c)(i)

- contains at least one carbon - carbon double bond

6(c)(ii)

- $\text{C}_3\text{H}_6\text{Br}_2$

6(d)(i)

- CH

6(d)(ii)

- addition

6(d)(iii)

- only two carbon atoms joined by a single bond and two additional bonds on each carbon
- correct repeat unit including two extension bonds

Easy explanation / reminder

- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

2025 Feb/March Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 14 | Answer source: 0620_m25_ms_42.pdf

Extra topic(s): Topic 3: Stoichiometry

Answer points

6(a)

- (have a) general formula of C_nH_{2n}
- (have a) $\text{C}=\text{C}$ functional group

6(b)

- compounds made of C and H only

6(c)

- $2\text{C}_3\text{H}_6 + 9\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$
- $\text{CO}_2 + \text{H}_2\text{O}$ as only products
- correct equation

6(d)

- CH_2

6(e)

- but-2-ene

6(f)

- displayed formula of but-1-ene

6(g)(i)

- only one product is formed

6(g)(ii)

- orange to colourless

6(g)(iii)

- 2,3-dibromobutane

6(h)(i)

- displayed formula of ethanoic acid
- February/March 2025

6(h)(ii)

- ethanoic acid

6(h)(iii)

- $\text{CH}_3\text{CH}=\text{CHCH}_3 + 2\text{O}_2 \rightarrow 2\text{CH}_3\text{COOH}$

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2025 May/June Paper 41 Question 1

Main topic: Topic 11: Organic chemistry | Marks: 10 | Answer source: 0620_s25_ms_41.pdf

Extra topic(s): Topic 9: Metals, Topic 2: Atoms, elements and compounds, Topic 10: Chemistry of the environment

Answer points**1(a)**

- calcium oxide

1(b)

- graphite

1(c)

- ethanol

1(d)

- methane

1(e)

- bauxite

1(f)

- sulfur dioxide

1(g)

- propane

1(h)

- calcium oxide

1(i)

- oxygen

1(j)

- propene

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2025 May/June Paper 41 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 18 | Answer source: 0620_s25_ms_41.pdf

Answer points**5(a)**

- concentration is no longer changing

- rates of forward reaction and of the reverse reaction are equal

5(b)(i)

- rate decreases

- yield of methanol decreases

5(b)(ii)

- no change(1) increases (1)

5(b)(iii)

- exothermic

5(b)(iv)

- copper

- transition element

5(c)(i)

- any 2 from:

- -

- same or similar chemical properties or reactions

- -

- (same) general formula

- -
- (consecutive members) differ by CH₂
- -
- same functional group
- -
- physical properties OR
- example of a physical property
- e.g. melting point /
- boiling point /
- volatility
- vary in predictable manner or show trends or gradually change

5(c)(ii)

- C₅H₁₂O

5(d)(i)

- displayed formula of ester linkage, i.e.
- H or C
- completely correct displayed formula of methyl ethanoate or ethyl methanoate
- i.e.

5(d)(ii)

- butan-1-ol
- ethanoic acid

5(e)

- C 64.87 / 12
- H 13.51 / 1
- O 21.62 / 16
- OR
- 5.41 13.51 1.35
- C₄H₁₀O

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2025 May/June Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 18 | Answer source: 0620_s25_ms_42.pdf

Answer points

6(a)

- homologous series

6(b)

6(c)

- C₂H₄ + 3O₂ -> 2CO₂ + 2H₂O
- CO₂ + H₂O
- correct equation

6(d)(i)

- decreases
- decreases
- decreases
- no change

6(d)(ii)

- kinetic energy of particles increases
- frequency of collisions between particles increases
- higher percentage / proportion / fraction of collisions / particles have energy greater than / equal to activation energy

6(e)(i)

- C₅H₆O₄

6(e)(ii)

- it has a carbon-carbon bond which is not a single bond

6(e)(iii)

- one C - C single bond with continuation bonds

- correct structure

6(e)(iv)

- compound B has two carboxylic acid groups

6(e)(v)

- (mol of NaOH reacting) = 0.2(00)
- volume = 1000 $\rightarrow$ 0.2(00) / 0.250 = 800 cm³

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2025 May/June Paper 43 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 18 | Answer source: 0620_s25_ms_43.pdf

Extra topic(s): Topic 6: Chemical reactions

Answer points

5(a)

- the rate of forward reaction is equal to the rate of the reverse reaction
- concentrations of reactants and products are no longer changing

5(b)(i)

- rate decreases
- yield of ethanoic acid decreases

5(b)(ii)

- no
- change
-
- increases
-
- increases

5(b)(iii)

- cobalt
- (it is a) transition element

5(c)(i)

- methanoic acid

5(c)(ii)

- C₄H₈O₂

5(d)(i)

- displayed formula of any ester linkage
- displayed formula of methyl methanoate

5(d)(ii)

- butanoic acid
- propan-1-ol

5(e)

-
- C 58.82 / 12
- H 9.80 / 1
- O 31.38 / 16
- OR
- 4.90:9.80:1.96
-
- C 4.90 / 1.96
- H 9.80 / 1.96
- O 1.96 / 1.96
- OR
- 2.5:5:1
- C₅H₁₀O₂

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2025 Oct/Nov Paper 41 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 20 | Answer source: 0620_w25_ms_41.pdf

Answer points

5(a)(i)

- $C_nH_{2n+1}OH$

5(a)(ii)

- (same) functional group

5(b)(i)

- 25-35

- 300

- yeast

- acid

- absence

- of oxygen

- 60 atm

5(b)(ii)

- $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

- $C_6H_{12}O_6$ as the only reactant on left

- correct equation

5(b)(iii)

- $C_2H_4 + H_2O \rightarrow C_2H_5OH$

- both $C_2H_4 + H_2O$ only on left

- correct equation

5(c)(i)

- $C_4H_{10}O_2$

5(c)(ii)

5(d)(i)

- $C_2H_3O_2$

5(d)(ii)

- hydrogen

5(e)(i)

- any one correct fully displayed ester link between any two blocks

- three fully displayed inter-block ester links with correct orientation

- fully correct diagram showing two complete repeat units of polyester including continuation bonds

5(e)(ii)

- condensation (polymerisation)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2025 Oct/Nov Paper 42 Question 5

Main topic: Topic 11: Organic chemistry | Marks: 12 | Answer source: 0620_w25_ms_42.pdf

Answer points

5(a)

- (have a) general formula of C_nH_{2n+2}

- differ from one member to the next by a $-CH_2-$ unit

5(b)(i)

- increases

5(b)(ii)

- physical property

- change as chain length increases

5(c)

- all carbon-carbon bonds are single (bonds)

5(d)(i)

- ultraviolet light

5(d)(ii)

- $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$
- 1-chloropropane
- $\text{CH}_3\text{CHClCH}_3$
- 2-chloropropane

5(d)(iii)

- (compounds with) the same molecular formula but different structural formulae or structure

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.*

2025 Oct/Nov Paper 42 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 8 | Answer source: 0620_w25_ms_42.pdf

Answer points

6(a)

- correctly circled ester link (including 2 C atoms and 2 O atoms)

6(b)

- ethyl butanoate

6(c)

- $\text{C}_3\text{H}_6\text{O}$

6(d)(i)

- butanoic acid
- ethanol

6(d)(ii)

- acid

6(e)

6(f)

- $2\text{C}_5\text{H}_{10}\text{O}_2 + 13\text{O}_2 \rightarrow 10\text{CO}_2 + 10\text{H}_2\text{O}$

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2025 Oct/Nov Paper 43 Question 6

Main topic: Topic 11: Organic chemistry | Marks: 17 | Answer source: 0620_w25_ms_43.pdf

Answer points

6(a)

- $\text{C } 50(.00) \div 12$
- $\text{H } 5.56 \div 1$
- $\text{O } 44.44 \div 16$
- OR
- $4.17:5.56:2.78$
- fractions shown dividing all by smallest
- OR
- $1.5:2:1$
- OR
- $3:4:2$
- $\text{C}_3\text{H}_4\text{O}_2$

6(b)

- $\text{C}_4\text{H}_4\text{O}_4$

6(c)(i)

- water

6(c)(ii)

- acid

6(c)(iii)

- propanoic acid
- butan-1-ol

6(d)(i)

- addition

6(d)(ii)

- polymers are of indefinite length

6(d)(iii)

- 3 / three

6(d)(iv)

- C = C
- displayed formula of but-2-ene
- OR

6(e)(i)

- -NH₂ displayed
- -COOH displayed
- functional groups can be on either side

6(e)(ii)

- one circle around either amide linkage

6(e)(iii)

- proteins

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*